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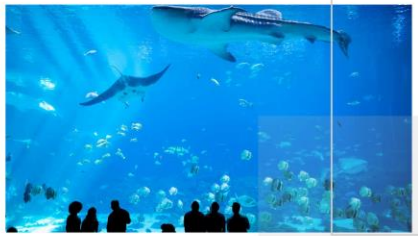

Estimation of Dependence and Marginal Distributions in Dependently Censored Survival Data Using the Generalized Gamma Distribution

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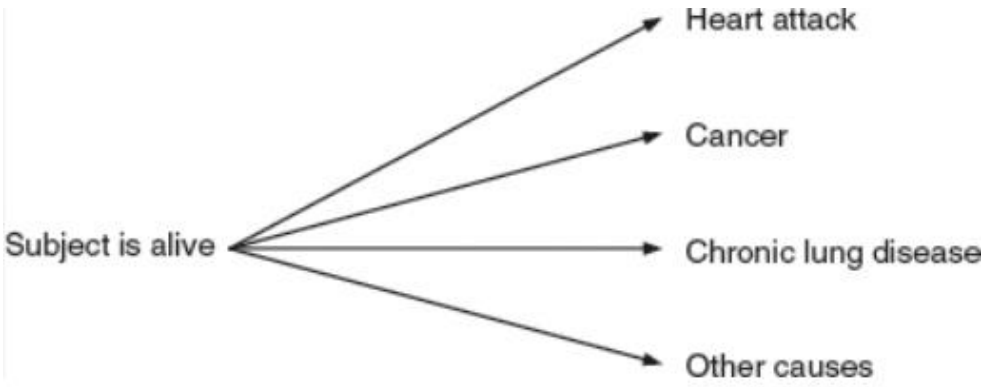
1. Background and Objectives

“Dependent” Competing Risks or Censoring in Survival Analysis

- **Dependence (Correlation) may exist between different causes of death**



- **In a randomized controlled trial (RCT), dependence may exist between the overall survival (OS) time and the time to patient dropouts due to adverse drug effects or deteriorating health**

Non-identifiability in competing risks

1. **Definition of identifiability**

   a. Let **T:** time-to-event of interest and **C:** time to any other competing event, where only the minimum of the two is observed and never both

   b. The observable data is **X = min(T, C) and δ = I(T ≤ C)**

   c. For the PDF f of the observables (X, δ) with parameters vector θ, θ is **identifiable** if any given θ uniquely determines the density f of (X, δ), i.e. **if** $\mathbf{f_{X,\delta,\theta_1}} \equiv \mathbf{f_{X,\delta,\theta_2}}$ **then** $\boldsymbol{\theta_1} = \boldsymbol{\theta_2}$ is satisfied

2. **The non-identifiability of competing risks (Tsiatis, 1975)**

   a. There is no way to **non-parametrically** distinguish independence from dependence of T and C

   b. Additional information regarding the dependence structure is needed. Here, we emphasize the use of **parametric** copulas and marginal distributions **for dependence modeling**

How to estimate dependence?

Correlated by some ρ_0

T	C	$I(T \leq C)$
t_1	missing (·)	1
t_2	missing (·)	1
missing (·)	c_3	0
·	·	·
·	·	·
·	·	·
t_{n-1}	missing (·)	1
missing (·)	c_n	0

1. **Why correlation estimation is difficult**
   a. Column-wise: **Only min(T, C) observed and never both**
   b. Row-wise: Different subjects, so independent of each other
2. **MLE may be problematic (under certain situations)**
   a. Maximum likelihood estimation (MLE) of the correlation parameter together with the marginal hazards parameters results in a large variability of the estimates due to the likelihood function having little information for the correlation *(Chen, 2010; Michimae & Emura, 2022)*
   b. Under maximum likelihood inference, the dependency parameter needs to be previously defined *(Schneider, 2023)*

Previous literature

1. **Zheng and Klein (1994, 1995)**
   a. **Assuming** that the strength and functional form of **the correlation** between T and C are **known** by a copula function and its correlation parameter, **the marginal distributions** of T and C **are identifiable** → The "copula-graphic (CG)" estimator
2. **Chen (2010), Emura and Chen (2016, 2018)**
   a. Extended the assumed copula approach to a general **semi-parametric regression** setting
3. **Deresa and Van Keilegom (2019~2023), Czado and Van Keilegom (2022)**
   a. Provided "parametric" identifiability of several widely used copulas and marginal distributions
   b. Assuming certain parametric copulas and marginal distributions, **the copula correlation parameter can be estimated** with likelihood-based methods (MLE)

Study objectives

1. Explicitly **estimate the correlation** between T (time to the event of interest) and C (time to other competing events or dependent censoring)
2. Use the estimated correlation between T and C to **estimate the marginal survival function of T** (time-to-event of interest)
3. Use the estimated correlation between T and C to **estimate the effect of a treatment or exposure on** the identified **marginal survival function of T**

2. Proposed Method

The "marginal" hazard function

1. **The well-known cause-specific and sub-distribution hazard functions (for event types 1, 2, …, K)**

   - $h_k^{c-s}(x) = \lim_{\Delta x \to 0} \frac{Pr(x \le X \le x+\Delta x, type=k \mid X \ge x)}{\Delta x}$

   - $h_k^{s-d}(x) = \lim_{\Delta x \to 0} \frac{Pr(x \le X \le x+\Delta x, type=k \mid X \ge x \cup (X < x \cap type \ne k))}{\Delta x}$

2. **The marginal hazard function (for the event of interest type k)**

   - $h_k^{marg}(x) = \lim_{\Delta x \to 0} \frac{Pr(x \le T_k \le x+\Delta x \mid T_k \ge x)}{\Delta x}$

   a. All other event types are removed or eliminated, and only the single type k is available to occur

   b. Note that **the cause-specific hazard is also a marginal hazard**, with the independence copula assumed to be its dependence structure

Copulas for dependency modeling

1. **The copula function**
   a. A **multivariate CDF** for which the marginal distribution of each random variable is Uniform(0, 1)
   b. For a random vector (X_1, X_2, …, X_K) of competing event survival times, the probability integral transform is applied to each marginal distribution to write the joint CDF as

   $Pr(X_1 \le x_1, X_2 \le x_2, \ldots, X_K \le x_K) = Pr(U_1 \le u_1, U_2 \le u_2, \ldots, U_K \le u_K) = C_\rho(u_1, u_2, \ldots, u_K)$,

   ρ **: copula correlation parameter**

2. **Some often used copulas**
   a. **Normal (Gaussian)** copula: $\frac{1}{2\pi\sqrt{1-\rho^2}}\int_{-\infty}^{\Phi_T^{-1}(u)}\int_{-\infty}^{\Phi_C^{-1}(v)} exp[-\frac{1}{2(1-\rho^2)}\{u^2 - 2\rho uv + v^2\}]\, dvdu,\ \rho \in [-1,1]$
   b. **Clayton** copula: $(u^{-\rho} + v^{-\rho} - 1)^{-\frac{1}{\rho}},\ \rho \in [-1,\infty)\backslash\{0\}$ → an Archimedean copula
   c. Other Archimedean copulas: Ex) **Frank** and **Gumbel** copulas
   d. **Independence** copula: $u \cdot v$ → no copula parameter, zero correlation

The generalized gamma (GG) distribution for time-to-events

1. **The versatile GG distribution**

   a. A generalization of the gamma distribution that nests other distributions widely used in parametric models of survival times: **exponential, Weibull, log-normal**, **gamma**, etc.

   b. For $X \sim GG(scale = \lambda, shape = \gamma, additional\ shape = \alpha)$,

   PDF $f_X(x) = [\alpha\lambda^{\gamma\alpha}x^{\gamma\alpha-1}\exp\{-(\lambda x)^{\alpha}\}]/\Gamma(\gamma)$, where x > 0 and $\Gamma(\gamma) = \int_0^{\infty} t^{\gamma-1} \cdot e^{-t}dt$

   ① For α=1: ***gamma(shape=γ, scale=λ)***

   ② For γ=1: ***Weibull(shape=α, scale=λ^{α})***

   ③ For α= γ=1: ***exponential(scale= λ)***

   ④ For $\gamma \to \infty$: ***log-normal($\mu = -log(\lambda) + log(\gamma)/\alpha, \sigma^2 = 1/\gamma\alpha^2$)***

2. **Here, we use the GG parameterization of (μ, σ, Q) by Prentice (1974) and Lawless (1980)**

Method-of-moments estimation (MME)

1. MME equates the first k sample moments to the corresponding k population moments and solves the system of equations simultaneously
2. Here, we equate the **first three sample moments (mean, variance, skewness)** of the observed X = min(T, C), δ = I(T ≤ C) with the three sub-distribution moments of the (hypothetical) bivariate GG distribution. We also utilize the **proportion of T observed in X**, $\mathrm{E}(\delta) = \Pr(\delta_i = 1,\ \mathrm{i}=1,\ 2,\ldots,\mathrm{n}) = \sum_{i=1}^{n} {}^{\delta_i}/_{n}$ for sample size n, as additional information

Proposed method: Estimable equations

1. **From the aforementioned:**

   a. **Marginal** hazard or survival function framework

   b. Parametric **copulas**, such as the Normal or Clayton, with copula **correlation parameter $\boldsymbol{\rho}$**

   c. **GG distributed** marginal survival times T and C

   d. **Method of moments** estimation (MME)

2. **Obtain seven estimable equations for seven parameters as**

   a. Equate $\sum_{i=1}^{n} {}^{\delta_i}/_{n}$, $\delta = \min(T, C)$ with that in the GG distribution

   b. Equate 1st to 3rd observed moments $\mathrm{E}(T \mid \delta = 1)$, $\mathrm{E}(C \mid \delta = 0)$, $\mathrm{E}(T^2 \mid \delta = 1)$, $\mathrm{E}(C^2 \mid \delta = 0)$, $\mathrm{E}(T^3 \mid \delta = 1)$, and $\mathrm{E}(C^3 \mid \delta = 0)$ with those in the GG distribution, respectively

   c. To estimate: correlation $\boldsymbol{\rho}$, $(\boldsymbol{\mu_T}, \boldsymbol{\sigma_T}, \boldsymbol{Q_T})$ of T **and** $(\boldsymbol{\mu_C}, \boldsymbol{\sigma_C}, \boldsymbol{Q_C})$ of C for bivariate GG distributed (T, C)

Proposed method: Correlation estimation algorithm

- **An estimation algorithm is thus conducted as:**

  I. **Starting from initial** parameters $\theta^{(0)} = (\mu_{\mathrm{T}}{}^{(0)}, \mu_{\mathrm{C}}{}^{(0)}, \sigma_{\mathrm{T}}^{2}{}^{(0)}, \sigma_{\mathrm{C}}^{2}{}^{(0)}, \mathrm{Q_T}{}^{(0)}, \mathrm{Q_C}{}^{(0)}, \rho^{(0)})$, generate a bivariate GG distributed random sample of size n (= size of given data (T, C))

  II. **Perform comparisons** of $\sum_{i=1}^{n} {}^{\delta_i}/_{n}, \delta = \min(T, C)$ and the 1st to 3rd observed moments to those in the generated bivariate GG distribution to minimize the mean loss (MSE, MAPE etc.) of an objective function. **Update** the bivariate GG parameters to those currently evaluated if mean loss decreases

  III. **Repeat** II. for a given number of times or until a convergence criterion is met

- The resulting bivariate GG parameters $\theta^{(k)} = (\mu_{\mathrm{T}}{}^{(\mathrm{k})}, \mu_{\mathrm{C}}{}^{(\mathrm{k})}, \sigma_{\mathrm{T}}^{2}{}^{(\mathrm{k})}, \sigma_{\mathrm{C}}^{2}{}^{(\mathrm{k})}, \mathrm{Q_T}{}^{(\mathrm{k})}, \mathrm{Q_C}{}^{(\mathrm{k})}, \rho^{(\mathrm{k})})$ after k iterations are those that match the dependently censored survival data (T, C), and specifically $\boldsymbol{\rho}^{(\mathrm{k})}$ **is the correlation estimate of (T, C)**

Proposed method: Estimation procedure flowchart

Obj. function

- **Objective function** to minimize aggregated sub-distribution moments comparison error between the given data (T, C) and its bivariate GG distribution counterpart
- $f_{obj}(\cdot) = (\sum_{j=1}^{7} w_j \times |\varepsilon_j|)$ for $\hat{\theta} = argmin_{\theta \in \Theta}(f_{obj}(\cdot))$

Param. space

- Calculate **lower and upper bounds of initial values** of the marginal distribution parameters using the **copula-graphic estimator** under **four candidate correlations** (none: 0, low: 0.3, moderate: 0.5, high: 0.8)

Global search

- Global search of the parameter space via **simulated annealing** under **four candidate correlations** (0, 0.3, 0.5 or 0.8). **Bootstrap aggregated voting (bagging)** determines which candidate correlation is selected

Local search

- Local **refinement of the correlation parameter** estimated in global search via gradient descent

Precision

- Calculate **confidence interval** of the correlation estimate using nonparametric bootstrap percentiles
- Calculate bootstrap confidence intervals' **coverage probability** in multiple generations of simulated (T, C) and corresponding bootstrap samples

3. Simulation Study

Simulation settings (1)

1. **The overall simulation configuration was based upon varying the three factors below**
   a. **Size of correlations**: Kendall's tau of 0, 0.3, 0.5, and 0.8
   b. **Marginal distributions** of survival times: Bivariate Weibull, Bivariate log-normal, T: Weibull and C: log-normal, Bivariate GG (general case), and T: log-normal, C: GG (general case)
   c. **Type of copulas**: Normal (Gaussian), Clayton, Frank, and Gumbel copulas
2. **Correlated survival times were generated by the "conditional CDF" method, utilizing the partial derivative of copula functions**

Simulation settings (2)

1. **The parameter settings for each of the marginal distributions followed those used in previous studies of copula modeling in competing risks or dependent censoring survival analysis**
   - The bivariate Weibull distribution's (shape, scale, Q) parameters were set as

     T ~ Weibull(0.63, 0.06, 1), C ~ Weibull(0.86, 0.04, 1), from Sorrell et al. (2022)
   - The bivariate log-normal distribution's (mean, SD, Q) of log(T) and log(C) were set as

     T ~ log-normal(2.2, 1.0, 0), C ~ log-normal(2.0, 0.25, 0), from Czado and Van Keilegom (2023)
   - The bivariate GG (general case) distribution's (μ, σ, Q) parameters were set as

     T ~ GG(4.23, 0.51, 0.31), C ~ GG(4.44, 0.29, 0.52), from Lawless (1980)
   - "Mixture" distributions of T: Weibull and C: log-normal, T: log-normal and C: GG (general case) each used the parameter settings among those specified above

Table 1-1: Correlation estimation results in 200 bootstrap samples of dependently censored survival data (T, C)

Marginal distributions	True Kendall's tau	Estimated Kendall's tau with proposed method			
		Point estimate	Bootstrap MAE	Bootstrap SE	Bootstrap 95% CI
Bivariate Weibull	0	0.111	0.060	0.070	(-0.094, 0.142)
	0.3	0.369	0.056	0.065	(0.175, 0.391)
	0.5	0.444	0.064	0.073	(0.406, 0.646)
	0.8	0.816	0.048	0.054	(0.701, 0.895)
Bivariate Log-Normal	0	-0.070	0.064	0.081	(-0.108, 0.207)
	0.3	0.266	0.022	0.045	(0.188, 0.381)
	0.5	0.483	0.023	0.032	(0.389, 0.552)
	0.8	0.825	0.059	0.071	(0.628, 0.912)
Weibull & Log-Normal	0	-0.060	0.060	0.070	(-0.092, 0.144)
	0.3	0.275	0.054	0.064	(0.162, 0.396)
	0.5	0.570	0.068	0.065	(0.420, 0.643)
	0.8	0.773	0.053	0.062	(0.673, 0.892)

Abbreviations: CI, confidence interval; **MAE**, mean absolute error; **SE**, standard error

Table 1-2: Correlation estimation results in 200 bootstrap samples of dependently censored survival data (T, C)

Marginal distributions	True Kendall's tau	Estimated Kendall's tau with proposed method			
		Point estimate	Bootstrap MAE	Bootstrap SE	Bootstrap 95% CI
Bivariate Generalized Gamma	0	0.061	0.060	0.069	(-0.095, 0.138)
	0.3	0.311	0.056	0.067	(0.157, 0.397)
	0.5	0.423	0.064	0.068	(0.415, 0.646)
	0.8	0.771	0.051	0.061	(0.671, 0.895)
Log-Normal & Generalized Gamma	0	0.036	0.065	0.068	(-0.089, 0.141)
	0.3	0.171	0.065	0.064	(0.166, 0.385)
	0.5	0.419	0.053	0.062	(0.404, 0.632)
	0.8	0.895	0.051	0.030	(0.788, 0.897)

Abbreviations: CI, confidence interval; **MAE**, mean absolute error; **SE**, standard error

Table 2: Correlation estimation results in 100 multiple runs of 100 bootstrap samples of dependently censored survival data (T, C)

Marginal distributions	True Kendall's tau	Estimated Kendall's tau with proposed method			
		Mean estimate	MAE	Empirical SE	CP
Bivariate Weibull	0	0.038	0.072	0.083	97
	0.3	0.271	0.067	0.078	97
	0.5	0.558	0.074	0.068	92
	0.8	0.793	0.058	0.076	97
Bivariate Log-Normal	0	0.013	0.063	0.083	98
	0.3	0.287	0.055	0.060	94
	0.5	0.492	0.054	0.049	96
	0.8	0.751	0.068	0.030	96
Weibull & Log-Normal	0	0.021	0.068	0.082	95
	0.3	0.286	0.070	0.090	97
	0.5	0.513	0.065	0.077	94
	0.8	0.761	0.065	0.079	96

Abbreviations: CP, coverage probability; **MAE**, mean absolute error; **SE**, standard error

Table 3: Correlation estimation results in 200 bootstrap samples of dependently censored survival data (T, C) under misspecified copulas, where the true copula is Normal (Gaussian)

Copula Applied	True Kendall's tau	Estimated Kendall's tau with proposed method			
		Median estimate	Bootstrap MAE	Bootstrap SE	Bootstrap 95% CI
Misspecified Clayton	0	0.026	0.064	0.071	(-0.092, 0.140)
	0.3	0.545	0.237	0.070	(0.410, 0.645)
	0.5	0.695	0.204	0.042	(0.651, 0.792)
	0.8	0.799	0.040	0.049	(0.707, 0.886)
Misspecified Frank	0	0.038	0.067	0.070	(-0.079, 0.141)
	0.3	0.039	0.268	0.071	(-0.094, 0.145)
	0.5	0.069	0.439	0.063	(-0.070, 0.145)
	0.8	0.743	0.067	0.061	(0.659, 0.874)
Misspecified Gumbel	0	0.027	0.062	0.069	(-0.094, 0.146)
	0.3	0.028	0.266	0.069	(-0.080, 0.147)
	0.5	0.281	0.226	0.066	(0.156, 0.388)
	0.8	0.712	0.082	0.056	(0.654, 0.859)

Abbreviations: CP, coverage probability; **MAE**, mean absolute error; **SE**, standard error

Figure 1. Boxplots of correlation estimates by different sample sizes

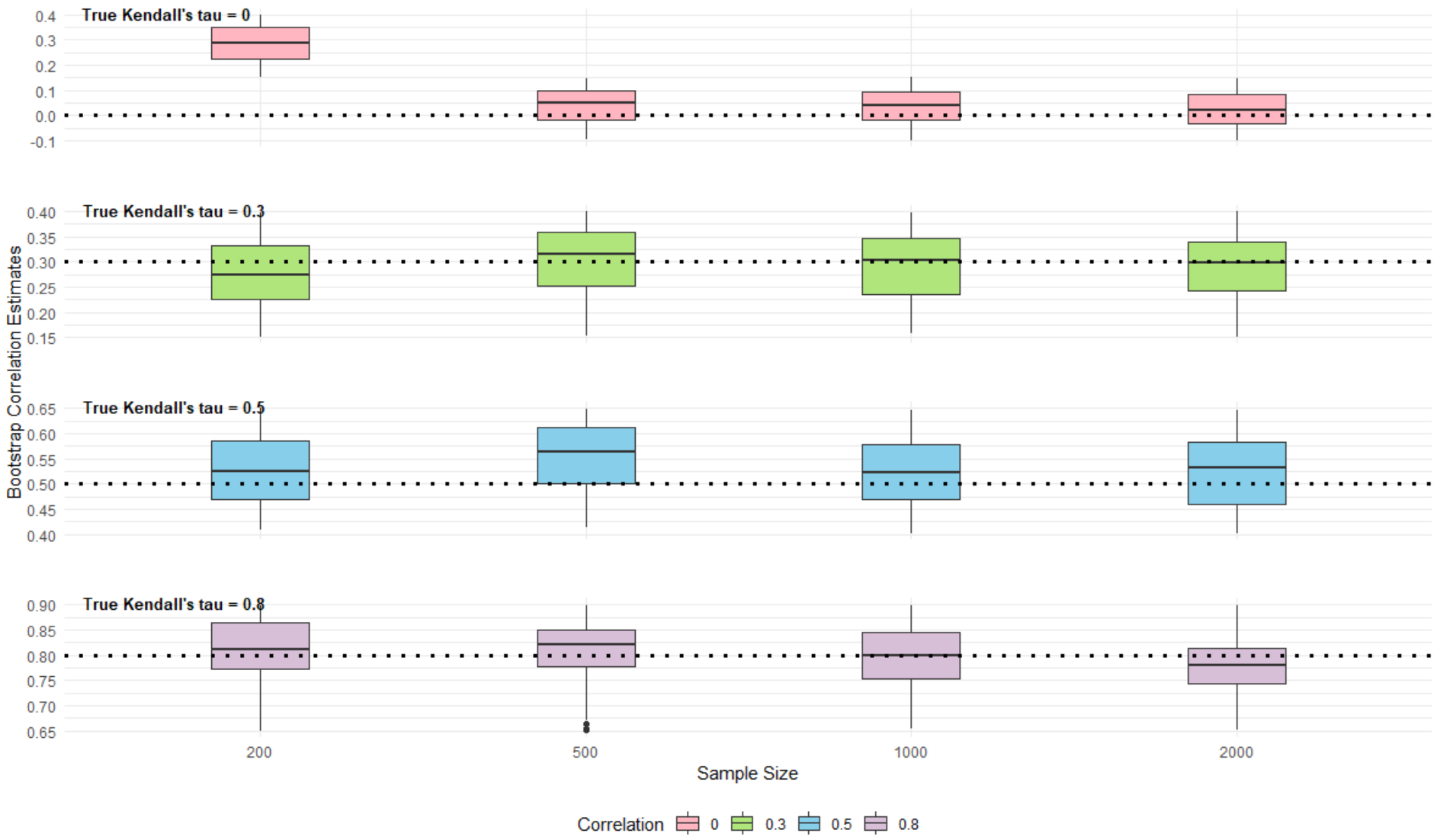
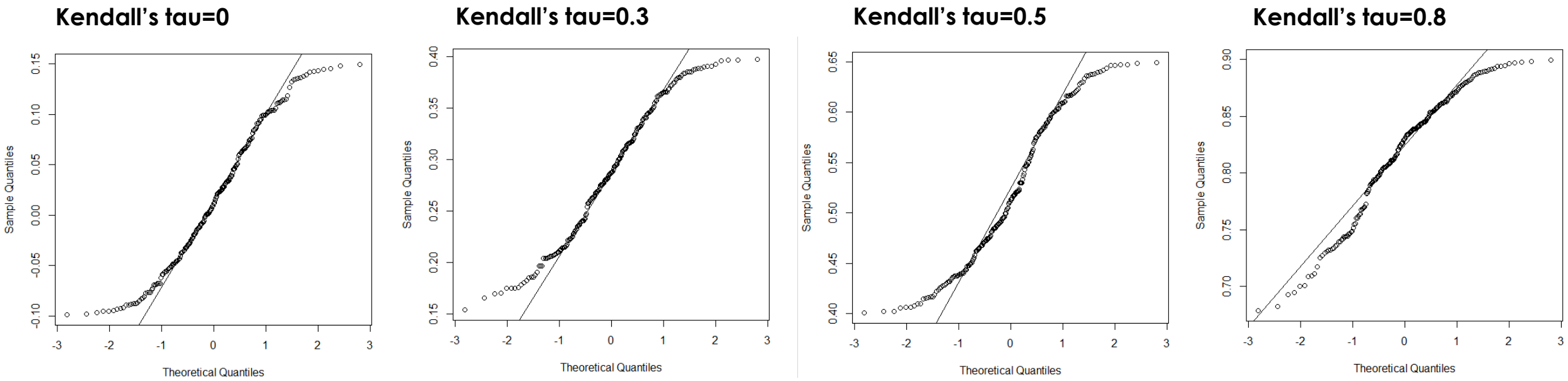


Figure 2. Q-Q plots of correlation estimates by different correlations (Kendall's tau: 0, 0.3, 0.5, and 0.8)



4. Real-world data application:

Patients with acute lymphoblastic leukemia (ALL)

Patients with ALL: Dataset description & preparation (1)

1. **A real-world dataset of 1,083 patients with ALL (after data-cleaning)**
   a. Available via the 'dynpred' package in R (European group for blood and marrow transplant)
   b. The patients received an allogeneic hematopoietic stem cell transplant (AHSCT) as a cure for ALL, after which acute graft vs. host disease (AGvHD) may occur
   c. Clinically relevant time-to-events: AGvHD incidence, relapse of ALL, all-cause death
2. **Additional prognostic variables**
   - Donor-recipient gender mismatch (yes/no), GvHD prophylaxis T-cell depletion (yes/no), Age at transplant (≤20, 20-40, >40), Year of transplant (1985-89, 1990-94, 1995-98)

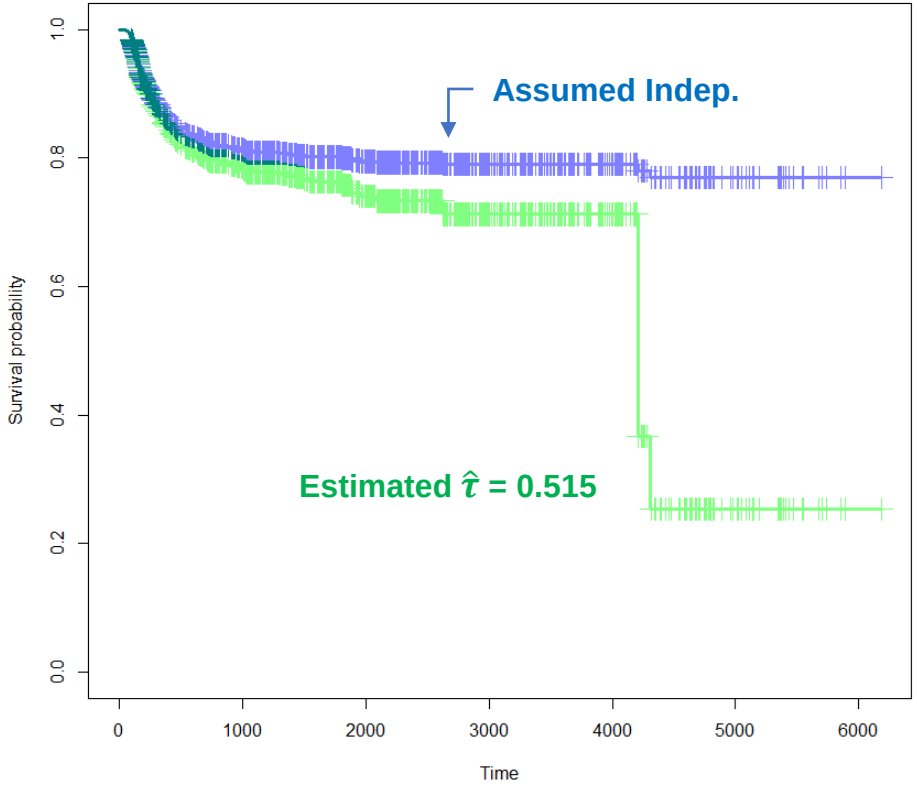
Patients with ALL: Dataset description & preparation (2)

1. **Research question: Does AGvHD have a beneficial (protective) effect on the time to relapse?**
   a. The occurrence of AGvHD, although life-threatening if very severe, is **also known to have a beneficial "graft vs. leukemia" effect**, where the transplanted healthy immune cells attack any residual cancerous blood cells that may remain in the host
   b. Important to consider the **dependence between** the time-to-relapse and time-to-death (before relapse), since it is very likely that patients with **longer/shorter time to relapse** would **also** have **longer/shorter time-to-death**
2. **Hence comes in our proposed method of correlation estimation between the survival endpoints to unbiasedly estimate the effect of AGvHD on the "marginal" hazard of relapse**

Table 4. Baseline characteristics of patients with ALL (N=1,083) in total and by incident acute graft vs. host disease (AGvHD) status

Variables	Total	AGvHD incidence		P-value
		Yes	No	
Overall N (%)	1,083 (100)	567 (52.4)	516 (47.6)	-
Age at transplant, N (%)				0.536
≤ 20	275 (25.4)	138 (24.4)	137 (26.6)	
20 - 40	533 (49.2)	278 (49.0)	255 (49.4)	
> 40	275 (25.4)	151 (26.6)	124 (24.0)	
Donor-recipient mismatch, N (%)				0.267
No mismatch	835 (77.1)	429 (75.7)	406 (78.7)	
Gender mismatch	248 (22.9)	138 (24.3)	110 (21.3)	
GvHD prevention, N (%)				0.066
No prophylaxis	870 (80.3)	468 (82.5)	402 (77.9)	
T-cell depletion prophylaxis	213 (19.7)	99 (17.5)	114 (22.1)	
Disease relapse, N (%)				0.327
Yes	204 (18.8)	100 (17.6)	104 (20.2)	
No	879 (81.2)	467 (82.4)	412 (79.8)	
Follow-up time (years), mean (SD)	4.91 (3.91)	4.83 (3.97)	5.01 (3.84)	0.453

Figure 3. Marginal survival curves of the time to relapse in patients with ALL under the proposed method's estimated correlation (in green), indep. (left, in blue), and an assumed correlation of 0.8 (right, in blue)



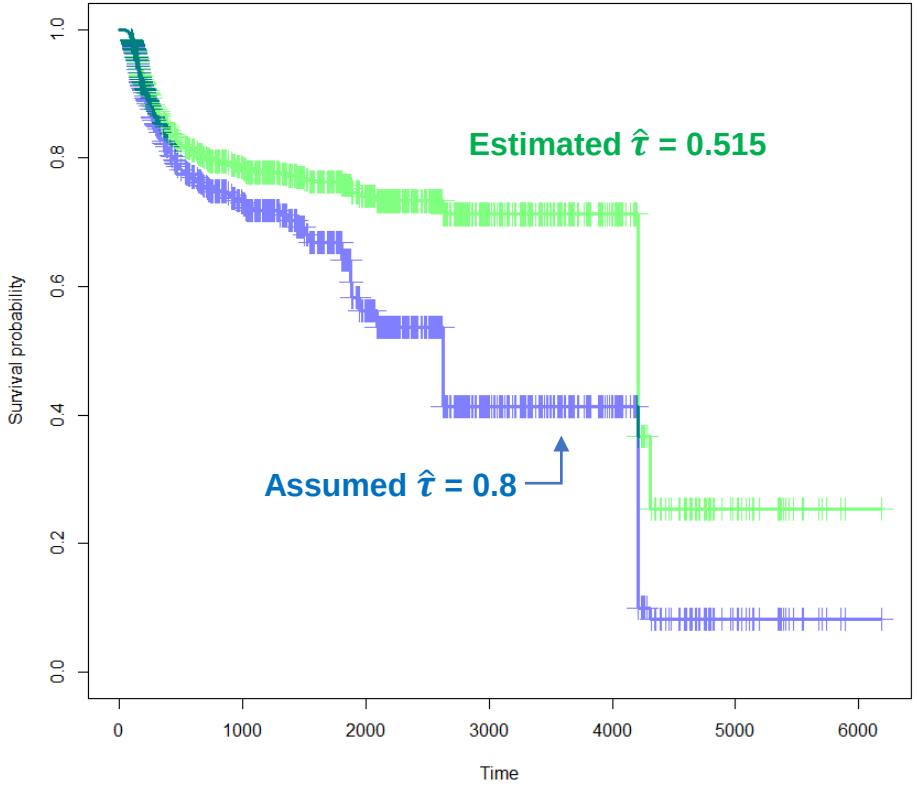


Table 5. Results of correlation estimation and subsequent regression coefficient estimation in a univariable Cox regression model of AGvHD status on the time to relapse in patients with ALL

	Proposed method of correlation estimation ($\hat{\tau}$ = 0.515 or 0.502)			**Cause-specific hazards (Assumed independence, $\hat{\tau}$ = 0)**			**Assumed correlation ($\hat{\tau}$ = 0.8)**		
Params.	Mean [a] estimate	Bootstrap SE [b]	One-sided P-value	Mean estimate	Bootstrap SE	One-sided P-value	Mean estimate	Bootstrap SE	One-sided P-value
β_T	**-0.128**	0.131	**0.082**	**-0.109**	0.141	**0.220**	**-0.105**	0.092	**0.063**
$\hat{\tau}$	0.515	0.075	<0.001	0	-	-	0.8	-	-
Params.	[c] Orig. data estimate	Orig. data SE	One-sided P-value	Orig. data estimate	Orig. data SE	One-sided P-value	Orig. data estimate	Orig. data SE	One-sided P-value
β_T	**-0.132**	0.132	**0.159**	**-0.112**	0.140	**0.211**	**-0.084**	0.089	**0.172**
$\hat{\tau}$	0.502	-	-	0	-	-	0.8	-	-

Abbreviations: AGvHD, acute graft vs. host disease; **ALL**, acute lymphoblastic leukemia; **Params.**, parameters for estimation; **SE**, standard error

[a] Mean value of 200 regression coefficients estimated from 200 bootstrap samples of the original ALL dataset

[b] Standard error (empirical standard deviation) of 200 regression coefficients estimated from the 200 bootstrap samples

[c] The single regression coefficient, SE, and P-value estimated from the original ALL dataset

5. Discussion and Conclusion

Points of discussion

1. **Some observations noted**
   a. Under the (μ, σ, Q) parameterization of the marginal distributions of T or C, the Q parameter tended to have the worst estimation performance in terms of mean percentage bias and coefficient of variation
   b. Although the gamma distribution's shape and scale parameters are easily estimated by its mean and variance (sample moments), this was not easily applicable as only the "sub-"distributions of T or C were observable
2. **Study limitations**
   a. The statistical properties of the proposed correlation estimator need to be well-examined
   b. Misspecification of the functional form of dependence (the type of copula function) resulted in biased correlation estimates, although the proposed estimator was still able to distinguish zero and high (= Kendall's tau of 0.8) correlation under all copula misspecifications
   c. Estimation performance needs to be evaluated for marginal survival distributions not covered by the generalized gamma, such as the log-logistic or Gompertz → *employ the generalized F distribution?*

Study conclusion

1. We proposed **a new method of estimating the correlation (strength of dependence)** in dependently censored survival data
2. The proposed method **does not require the presence of an informative covariate(s)**, and by utilizing **the generalized gamma distribution that encompasses the most widely used parametric distributions** in survival analysis, it is a more robust alternative compared to existing methods
3. The proposed method **may have further applications in drug efficacy RCTs and patient survival etiology studies**, where dependence between the survival endpoints exist more often than not

References

- A non-identifiability aspect of the problem of competing risks. Tsiatis (1975)
- Semiparametric marginal regression analysis for dependent competing risks under an assumed copula. Chen (2010)
- Likelihood Inference for Copula Models Based on Left-Truncated and Competing Risks Data from Field Studies. Michimae and Emura (2022)
- Clayton copula for survival data with dependent censoring: An application to a tuberculosis treatment adherence data. Schneider (2023)
- A self-consistent estimator of marginal survival functions based on dependent competing risk data and an assumed copula. Zheng and Klein (1994)
- Estimates of marginal survival for dependent competing risks based on an assumed copula. Zheng and Klein (1995)
- Gene selection for survival data under dependent censoring: A copula-based approach. Emura and Chen (2016)
- Flexible parametric model for survival data subject to dependent censoring. Deresa and Van Keilegom (2019)
- Dependent censoring based on parametric copulas. Czado and Van Keilegom (2023)
- Estimating the correlation between semi-competing risk survival endpoints. Sorrell et al. (2022)
- A log-gamma model and its maximum likelihood estimation. Prentice (1974)
- Inference in the generalized gamma and log-gamma distributions. Lawless (1980)

Thank you for your time and attention! ☺

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Have a blissful rest of the week!